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A network analysis of depression and anxiety symptoms among Chinese elderly living alone: based on the 2017–2018 Chinese Longitudinal Healthy Longevity Survey (CLHLS)

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Abstract

Background Elderly individuals living alone represent a vulnerable group with limited family support, making them more susceptible to mental health issues such as depression and anxiety. This study aims to construct a network model of depression and anxiety symptoms among older adults living alone, exploring the correlations and centrality of different symptoms. The goal is to identify core and bridging symptoms to inform clinical interventions.

Methods Using data from the 2018 Chinese Longitudinal Healthy Longevity Survey (CLHLS), this study constructed a network model of depression and anxiety symptoms among elderly individuals living alone. Depression and anxiety symptoms were assessed using the Center for Epidemiologic Studies Depression Scale-10 (CESD-10) and the Generalized Anxiety Disorder Scale-7 (GAD-7), respectively. A Gaussian Graphical Model (GGM) was employed to build the symptom network, and the Fruchterman-Reingold algorithm was used for visualization, with the thickness and color of the edges representing partial correlations between symptoms. To minimize spurious correlations, the Least Absolute Shrinkage and Selection Operator (LASSO) method was applied for regularization, and the optimal regularization parameters were selected using the Extended Bayesian Information Criterion (EBIC). We further calculated Expected Influence (EI) and Bridge Expected Influence (Bridge EI) to evaluate the importance of symptoms. Non-parametric bootstrap methods were used to assess the stability and accuracy of the network.

Results The Network centrality analysis revealed that GAD2 (Uncontrollable worry) and GAD4 (Trouble relaxing) exhibited the highest strength centrality (1.128 and 1.102, respectively), indicating their significant direct associations with other symptoms and their roles as core nodes in the anxiety symptom network. Other highly central nodes, such as GAD1 (Nervousness or anxiety) and GAD3 (Generalized worry), further underscore the dominance of anxiety symptoms in the overall network. Betweenness centrality results highlighted GAD1 (Nervousness or anxiety) and GAD2 (Uncontrollable worry) as critical bridge nodes facilitating information flow between different symptoms, while CESD3 (Feeling depressed) demonstrated a bridging role across modules. Weighted analyses further confirmed the central

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importance of GAD2 (Uncontrollable worry) and GAD4 (Trouble relaxing). Additionally, the analysis showed gender differences in the depression-anxiety networks of elderly individuals living alone.

Conclusion This study, through network analysis, uncovered the complex relationships between depression and anxiety symptoms among elderly individuals living alone, identifying GAD2 (Uncontrollable worry) and GAD4 (Trouble relaxing) as core symptoms. These findings provide essential insights for targeted interventions. Future research should explore intervention strategies for these symptoms to improve the mental health of elderly individuals living alone.

Keywords Elderly living alone, Depression and anxiety, Network analysis, Core symptoms

Introduction

Currently, the mental health issues of the elderly have garnered widespread attention in recent years, particularly concerning the depression and anxiety experienced by elderly individuals living alone [1]. As the trend of population aging intensifies in China, an increasing number of seniors are choosing or being compelled to live independently [2]. In fact, due to factors such as social support [3] and substance abuse [4], elderly individuals living alone often face more severe mental health issues, including anxiety, depression, feelings of loneliness, and a sense of worthlessness [5]. The lack of daily social interaction and support makes these seniors particularly susceptible to feelings of isolation, which are significant contributors to depression and anxiety [6, 7]. Research indicates that elderly individuals living alone are more likely to experience loneliness and social isolation compared to their counterparts who live with others [8]. This sense of loneliness not only affects their emotional state but may also lead to a deterioration in physical health, severely impacting both the psychological and physiological well-being of elderly individuals living independently [9].

A substantial body of research has examined the relationship between living alone and the prevalence of depression and anxiety. Studies indicate that among the elderly population in China, individuals living alone face a significantly higher risk of developing depression compared to their non-living-alone counterparts [10]. Moreover, living alone may serve as a straightforward yet effective predictive indicator for identifying high-risk groups for depression among the elderly [11]. A large-scale study conducted within the UK population has also confirmed that living alone is a high-risk factor for the onset of depression [12]. Anxiety, as one of the predominant psychological challenges faced by elderly individuals living in solitude, frequently coexists with depression. Research indicates that seniors residing alone often encounter anxiety stemming from factors such as the fear of isolation, physical deterioration, pervasive feelings of loneliness, and financial hardships [13]. A systematic review conducted by Ciuffreda et al. further suggests

that feelings of loneliness are intricately linked to anxiety and depressive symptoms among the elderly during the COVID-19 pandemic [14]. Moreover, due to communication barriers, self-neglect, and the societal stigma associated with mental disorders, elderly individuals living alone may experience significant delays in accessing healthcare services [15]. These factors collectively contribute to a deterioration in quality of life and an escalation in mortality rates among seniors living independently [16]. Consequently, the issues of depression and anxiety among elderly individuals living alone represent a multifaceted and pressing social concern. A thorough understanding of the intrinsic relationship between depression and anxiety within this demographic is essential for formulating effective prevention and intervention strategies.

Network analysis is an emerging data analysis method that represents the characteristics and information of a system in the form of a network. Networks consist of "nodes" and "edges" used to observe the interactions and associations between factors of interest. The network theory of mental disorders is based on network analysis methods [17]. It conceptualizes mental disorders as symptom networks, where causal relationships among symptoms and their dynamic interactions lead to the onset and development of mental disorders [18]. For instance, in treating individuals with depression, symptoms exhibit causal relationships, allowing for the prediction of dynamic changes in related symptoms through the alleviation of one symptom [19]. Network analysis of mental disorders enables the identification of core symptoms that are most closely related to surrounding symptoms and bridge symptoms that link different mental disorders. These critical symptoms are essential for maintaining the complex network and may play unique roles in diagnosis, prediction, and treatment [18, 20].

Despite some existing studies on depression and anxiety in elderly individuals living alone, several limitations remain [21–23]. Most research on mental disorders in the elderly has focused on anxiety or depression as overarching categories, neglecting the interrelations and mutual influences between clinical

symptoms, thereby failing to highlight the core symptoms within the multifaceted presentations of mental disorders. Many studies employ relatively traditional methods, such as basic quantitative surveys or case reports, without fully leveraging modern data analysis techniques [11, 24]. Network analysis specifically targeting anxiety and depression symptoms in elderly individuals living alone is scarce. Therefore, this study applies network analysis to uncover the relationships between depression and anxiety symptoms in elderly individuals living alone in China. By identifying core symptoms and bridge symptoms within the symptom network, this research aims to provide a comprehensive understanding of the complexity of depression and anxiety issues in this population, and explore potential intervention strategies.

Materials and methods

Study participants

This study utilizes data from the 2017–2018 wave of the Chinese Longitudinal Healthy Longevity Survey (CLHLS). The CLHLS was initiated in 1998 and has since conducted follow-up surveys in 2000, 2002, 2005, 2008–2009, 2011–2012, 2014, and 2017–2018 across 23 provinces in China, continuously recruiting new participants to maintain a consistent sample size [25]. The survey process is carried out by well-trained professionals using structured questionnaires covering demographic characteristics, socioeconomic status, daily living abilities, physical and mental health, cognitive function, lifestyle factors, and social support [26]. The CLHLS study received ethical approval from the Peking University Institutional Review Board (IRB00001052-13074), and written informed consent was obtained from each participant or their authorized representative. Participation in the CLHLS is voluntary, and respondents are free to withdraw from the study at any time without any consequences.

In this study, we focused on elderly individuals living alone, defined as those who do not have a spouse or any other family members living with them in the same household. These individuals were classified as "living alone" based on their self-reported living arrangements in the questionnaire. Participants below the age of 60 were excluded to ensure that the study sample conforms to the established definition of the elderly population. Furthermore, individuals with over 20% missing data on critical variables (such as CESD-10 and GAD-7) were also excluded from the analysis. Data from a total of 2028 elderly individuals living alone were included in this study.

Survey instruments

This study used the 10-item version of the Center for Epidemiologic Studies Depression Scale (CESD-10) to assess depression among elderly individuals living alone [27]. The scale has a Cronbach's α coefficient of 0.809, indicating good internal consistency. The CESD-10 encompasses ten items related to insomnia, feelings of sadness, hopelessness, and depression. Its validity and reliability have been extensively validated within the Chinese population, establishing it as a crucial instrument for assessing depressive symptoms among this demographic [28]. Scores range from 0 to 30, with higher scores indicating more severe depression. A CESD-10 score of ≥ 11 is considered indicative of depressive symptoms, while a score of < 10 suggests no significant depressive symptoms. To evaluate anxiety levels among elderly individuals living alone, this study employed the Generalized Anxiety Disorder 7-item scale (GAD-7) [29]. The GAD-7 comprises seven items that cover various aspects of generalized anxiety symptoms. Scores range from 0 to 21, with higher scores indicating more severe anxiety symptoms. A GAD-7 score > 5 is considered indicative of anxiety symptoms. The Chinese versions of both the CESD-10 and GAD-7 scales have been validated and shown to possess good reliability and validity [30]. These scales are widely used in assessing mental disorders among the elderly in China [31, 32].

Network analysis

In this study, we employed a Gaussian Graphical Model (GGM) to construct a network of depressive and anxiety symptoms among elderly individuals living alone [33]. We preprocessed the data, which included handling missing values and standardizing the scores. We visualized the symptom network using the Fruchterman-Reingold algorithm [34], where the thickness and color of the edges represent the partial correlations between symptoms. To eliminate spurious correlations, we applied the LASSO method for regularization and utilized the Extended Bayesian Information Criterion (EBIC) to select the optimal regularization parameter [35, 36]. The core symptoms of the network are primarily measured using centrality indices, with higher centrality scores indicating a more pivotal role of the symptoms. The centrality indices include strength centrality, bridge centrality, and closeness centrality, all of which serve to assess the significance of nodes within the network. Strength centrality refers to the sum of the absolute values of the connection strengths between a particular symptom and other symptoms. Closeness centrality measures the average shortest path length from a symptom to all other symptoms. Bridge centrality assesses the extent to

which a symptom acts as a bridge in the shortest paths between other symptoms within the network. A higher bridge centrality indicates that the symptom plays a more crucial role in connecting other symptoms, serving as a key mediator in the transmission of symptoms. We calculated the bridge expected influence to evaluate the bridging effect across symptom clusters. The barrat's strength and onnella's weighted measure are utilized in weighted network analysis to identify significant nodes within local networks. Additionally, we employed a non-parametric bootstrap method to evaluate the stability and accuracy of the network, identifying core and bridging symptoms within the domains of depression and anxiety.

Result

Baseline characteristics

A total of 2,028 older adults living alone were included in the analysis. The average age was 84.28 years ($SD=9.47$), with 775 male and 1,253 female. The overall mean score for the CESD-10 was 8.21 ($SD=4.73$), and the mean score for the GAD-7 was 1.61 ($SD=3.00$). The mean values and standard deviations for all CESD-10 and GAD-7 items are presented in Table 1.

Symptom network analysis of depression and anxiety among older adults living alone

The network structure of depression and anxiety symptoms among older adults living alone is illustrated in

Table 1 Basic information of depression-anxiety symptom network nodes among older adults living alone

Label	Items	Mean	SD	Predictability(R^2)
CESD1	Feeling bothered	0.34	0.64	0.396
CESD2	Trouble focus	0.64	0.82	0.199
CESD3	Feeling depressed	0.32	0.61	0.48
CESD4	Trouble doing anything	0.81	0.89	0.341
CESD5	Hopelessness	1.57	1.09	0.239
CESD6	Felt fearful	0.26	0.56	0.316
CESD7	Lack of happiness	1.81	1.17	0.231
CESD8	Loneliness	0.66	0.85	0.324
CESD9	Inability to get going	0.23	0.55	0.317
CESD10	Sleep quality	1.57	0.93	0.161
GAD1	Nervousness or anxiety	0.35	0.59	0.605
GAD2	Uncontrollable worry	0.25	0.54	0.701
GAD3	Generalized worry	0.29	0.58	0.662
GAD4	Trouble relaxing	0.21	0.50	0.669
GAD5	Restlessness	0.18	0.48	0.614
GAD6	Irritability	0.18	0.46	0.561
GAD7	Fear of horrible events	0.15	0.44	0.543

Note: SD standard deviation, CESD-10 Center for Epidemiologic Studies Depression scale – 10, GAD-7 Generalized Anxiety Disorder 7-item scale

Fig. 1. A circular pie chart is used to represent node predictability, with an average predictability of 0.433 (ranging from 0.199 to 0.701, see Table 1). Within the depression symptom network, the strongest associations were observed between CESD1 (Feeling bothered) and CESD3 (Feeling depressed), followed by CESD2 (Trouble focusing) and CESD4 (Trouble doing anything), as well as CESD5 (Hopelessness) and CESD7 (Lack of happiness). In the anxiety symptom network, the strongest associations were found between GAD1 (Nervousness or anxiety) and GAD2 (Uncontrollable worry), followed by GAD2 (Uncontrollable worry) and GAD3 (Generalized worry), as well as GAD4 (Trouble relaxing) and GAD5 (Restlessness).

In the analysis of network centrality (results shown in Fig. 2 and Table 2), Strength measures the total direct connections of a node to other nodes, reflecting its overall importance within the network. GAD2 (Uncontrollable worry) and GAD4 (Trouble relaxing) exhibited the highest strength centrality (1.126 and 1.106, respectively), indicating these symptoms have the strongest direct associations with other symptoms and are likely core nodes in the anxiety symptom network. Other high-strength nodes, such as GAD1 (Nervousness or anxiety) and GAD3 (Generalized worry), demonstrated extensive connectivity, further underscoring the central role of anxiety symptoms within the network. Betweenness centrality, which quantifies a node's role as a bridge within the shortest paths between other nodes, reflects its ability to regulate the flow of information. GAD1 (Nervousness or anxiety) and GAD2 (Uncontrollable worry) had the highest betweenness values (52 and 46, respectively), highlighting their prominent roles as critical hub symptoms. CESD3 (Feeling depressed) also showed a relatively high betweenness value (36), suggesting its significant role in connecting depressive and anxiety symptom clusters. The closeness centrality values for all nodes were relatively similar (approximately 0.003–0.004), indicating a uniformly connected network where symptoms are equidistant from each other, reflecting robust global connectivity within the network.

In the analysis of weighted measures (results shown in Table 2), Barrat's Strength, which incorporates edge weights to assess the total connectivity of a node, provides a more nuanced understanding of symptom importance. Consistent with the Strength findings, GAD2 (Uncontrollable worry) and GAD4 (Trouble relaxing) had the highest Barrat's Strength values, further supporting their roles as central nodes in the anxiety network. Additionally, CESD3 (Feeling depressed) displayed a high weighted strength (1.023), highlighting its influence within the depression symptom network. Onnella's Weighted Measure, which integrates both local

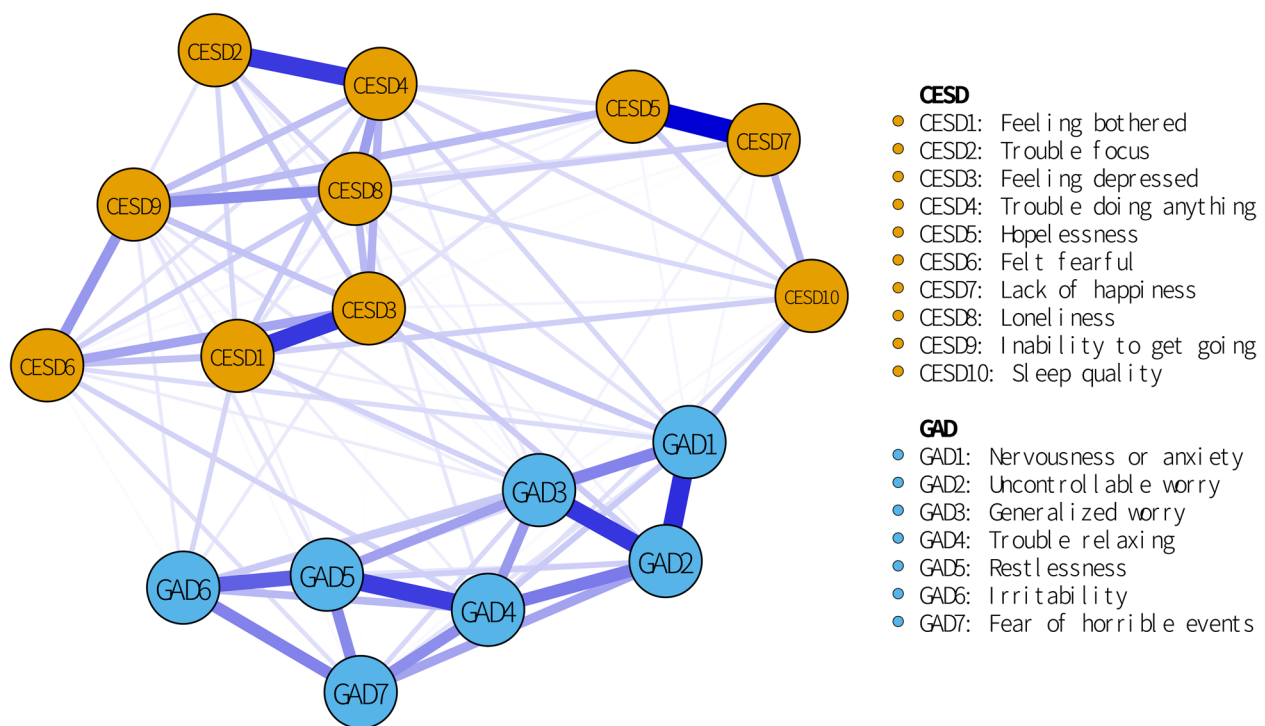


Fig. 1 Symptom network of depression and anxiety in older adults living alone. (Nodes: Represent symptoms of depression and anxiety. Edges: Indicate partial correlations between symptoms. Solid blue edges denote positive correlations, while dashed red edges indicate negative correlations. The thickness of the edges corresponds to the strength of the correlation.)

weighted connectivity and global network properties, was employed to evaluate node significance within the weighted network. GAD2 (Uncontrollable worry) and GAD4 (Trouble relaxing) again stood out with prominent values (0.794 and 0.758, respectively). Notably, GAD5 (Restlessness) and CESD2 (Trouble focus) also exhibited relatively high Onnela values (0.903 and 0.904, respectively), suggesting these symptoms may have localized importance and play key roles within specific substructures of the network.

Bridge symptoms analysis in the depression-anxiety network

Bridge symptoms are those with the highest bridge strength centrality, representing the most influential symptoms within the network. Our analysis of bridge symptoms in the depression-anxiety network (see figure) revealed that GAD1 (Nervousness or anxiety, Bridge Expected Influence=3.346) had the highest bridge strength, followed by GAD2 (Uncontrollable worry, Bridge Expected Influence=3.232), GAD3 (Generalized worry, Bridge Expected Influence=3.230), CESD3 (Feeling depressed, Bridge Expected Influence=3.094), and GAD4 (Trouble

relaxing, Bridge Expected Influence=3.036). These symptoms play a crucial role in linking depression and anxiety within the network (Fig. 3).

Gender differences in the depression-anxiety network among elderly living alone

The depression-anxiety symptom networks for elderly individuals living alone, stratified by gender, are shown in Fig. 4. In the network for males (Table 3 and Supplementary Fig. 1), the symptoms with the highest centrality were CESD3 (Feeling depressed) and CESD4 (Trouble doing anything) for depression, and GAD4 (Trouble relaxing) and GAD2 (Uncontrollable worry) for anxiety. In the female network (Table 3 and Supplementary Fig. 2), the symptoms with the highest centrality for depression were CESD3 (Feeling depressed) and CESD8 (Loneliness), while for anxiety, the symptoms GAD2 (Uncontrollable worry) and GAD4 (Trouble relaxing) had the highest centrality. The Network Comparison Test indicated a significant difference between the two network models (Test statistic M: 0.243, $P=0.027$), although no significant difference was found in strength centrality between the two networks ($P=0.716$).

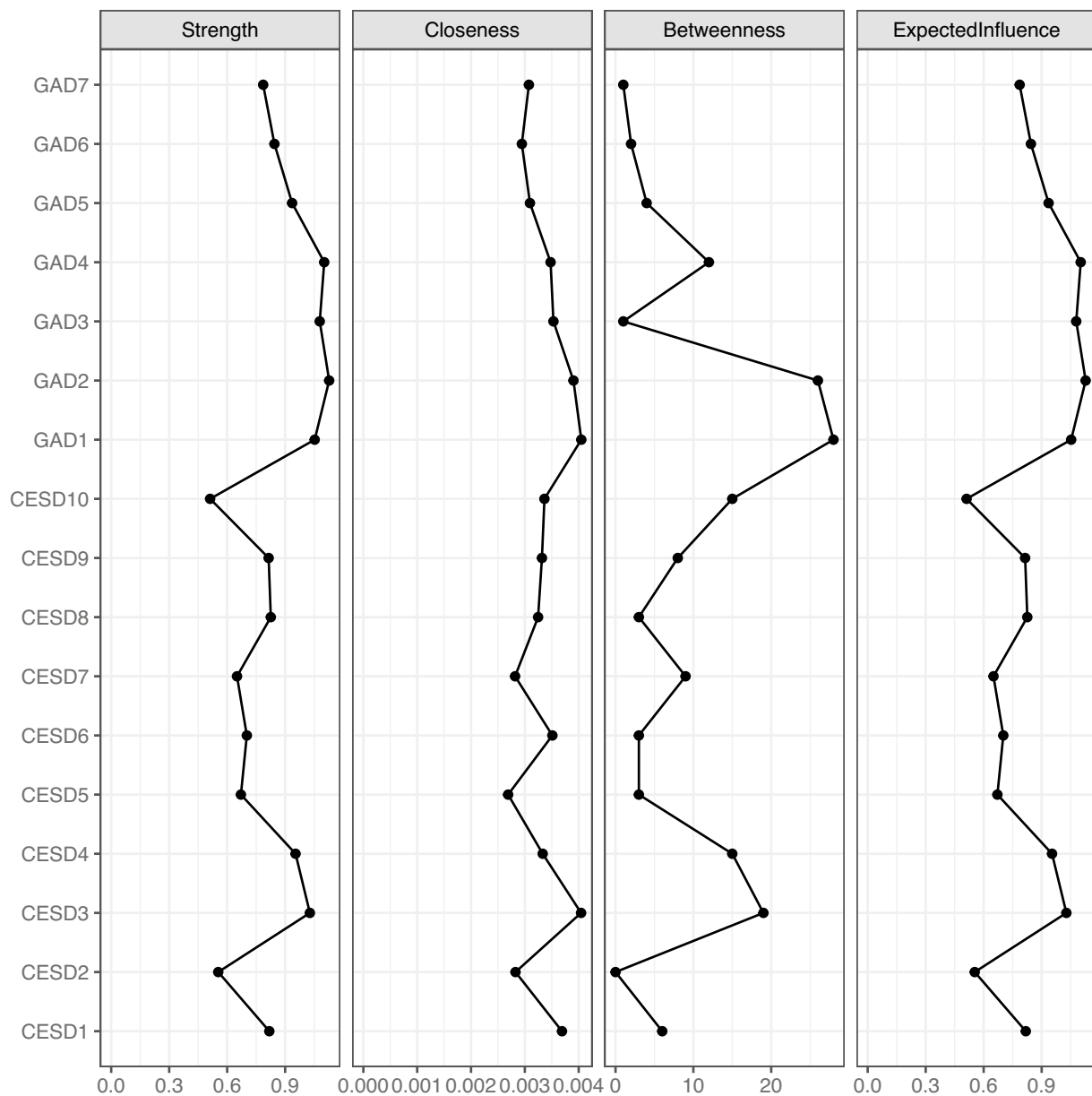


Fig. 2 Centrality measures of depression and anxiety symptoms in older adults living alone

Network stability and accuracy

We conducted stability analysis, accuracy assessment, and difference testing on the depression-anxiety symptom network of elderly individuals living alone. The stability of centrality indices was evaluated (as shown in Fig. 5A), revealing correlation stability coefficients of 0.75, 0.672, and 0.128 for strength, closeness, and betweenness centrality, respectively. These results indicate that the stability of strength and closeness centrality indices is relatively good, while the stability of betweenness centrality is poor. The accuracy of edge weights in

the symptom network was analyzed using bootstrap methods. The results presented in Fig. 5B show that the 95% confidence intervals for edge weights are narrow, indicating accurate assessment of edge weights in this study. The results of strength and edge weight difference testing can be found in Supplementary Figs. 3–4.

Discussion

This study investigates the interrelationship between anxiety and depression symptoms among elderly individuals living alone in China through the lens of network

Table 2 Descriptive statistics of centrality indices

Label	Items	Strength	Closeness	Betweenness	Barrat	Onnela
CESD1	Feeling bothered	0.82	0.004	12	0.82	0.782
CESD2	Trouble focus	0.551	0.003	0	0.551	0.904
CESD3	Feeling depressed	1.023	0.004	36	1.023	0.751
CESD4	Trouble doing anything	0.954	0.003	30	0.954	0.815
CESD5	Hopelessness	0.669	0.003	6	0.669	0.845
CESD6	Felt fearful	0.697	0.004	16	0.697	0.775
CESD7	Lack of happiness	0.646	0.003	16	0.646	0.766
CESD8	Loneliness	0.838	0.003	6	0.838	0.763
CESD9	Inability to get going	0.815	0.003	18	0.815	0.689
CESD10	Sleep quality	0.508	0.003	26	0.508	0.678
GAD1	Nervousness or anxiety	1.05	0.004	52	1.05	0.711
GAD2	Uncontrollable worry	1.126	0.004	46	1.126	0.794
GAD3	Generalized worry	1.075	0.004	2	1.075	0.696
GAD4	Trouble relaxing	1.106	0.003	30	1.106	0.758
GAD5	Restlessness	0.936	0.003	6	0.936	0.903
GAD6	Irritability	0.844	0.003	4	0.844	0.831
GAD7	Fear of horrible events	0.789	0.003	2	0.789	0.794

Note: The numbers in the table represent standardized Z values

theory, thereby offering a novel theoretical framework for comprehending the complexities of mental health in older adults. To the best of our knowledge, this represents the first systematic inquiry focused on this specific demographic. Previous research has indicated that the core driving factors of comorbid depression and anxiety networks among disabled elderly individuals [37], those with hypertension [38], and those with diabetes [39] in China are primarily depression-related symptoms (CESD3:Feeling Depressed). Our research emphasizes the critical role of anxiety-related symptoms, particularly GAD2 (Uncontrollable Worry) and GAD4 (Difficulty Relaxing), as core components of the symptom network. Their central position, both in terms of direct connectivity and weighted importance, indicates that these symptoms are integral to maintaining the anxiety-depression network among elderly individuals living alone and can serve as primary targets for therapeutic interventions. Furthermore, the bridging role of CESD3 (Feeling Depressed) highlights the interconnected nature of anxiety and depression, underscoring the importance of addressing the interactions between symptoms across disorders.

It has been reported that bridge symptoms may play a crucial role in the maintenance and progression of comorbid mental disorders and could be key targets for prevention and treatment [40]. The study delineates pivotal bridging symptoms, including GAD1 (Nervousness or anxiety), GAD2 (Uncontrollable worry), GAD3 (Generalized worry), and CESD3 (Feeling depressed),

which may form an interrelated core symptom cluster that collectively impacts the mental health and quality of life of elderly individuals residing alone. According to the DSM-5, GAD1 (Nervousness or anxiety) and GAD2 (Uncontrollable worry) are prominent features of anxiety disorders [41]. Previous research also indicates that during the COVID-19 pandemic, Uncontrollable worry was identified as a key symptom of anxiety and was closely associated with Nervousness or anxiety [42]. The role of Nervousness as a bridge symptom between anxiety and depression comorbidity has also been highlighted in studies on hypertensive elderly in China [43]. Research suggests that emotional support from other family members or peers can alleviate anxiety among the elderly [38]. Due to a lack of interaction with their children, elderly individuals living alone are more likely to experience psychological changes such as worry and nervousness. Among the various depressive symptoms, CESD-3 (Feeling depressed) emerged as a critical bridge symptom. Although the CESD scale is less commonly used in studies on anxiety and depression comorbidity, similar conclusions have been found in related research. The PHQ-9 is a brief scale used to assess the severity of depression, consisting of nine items, each representing a depressive symptom [44]. PHQ2 (Feeling down, depressed, or hopeless) is similar to CESD-3 (Feeling depressed) in its expression. Bian et al. found that PHQ2 was a bridge symptom in the network model of anxiety, depression, and personal control among elderly individuals living in the community [45]. These findings indicate that GAD1

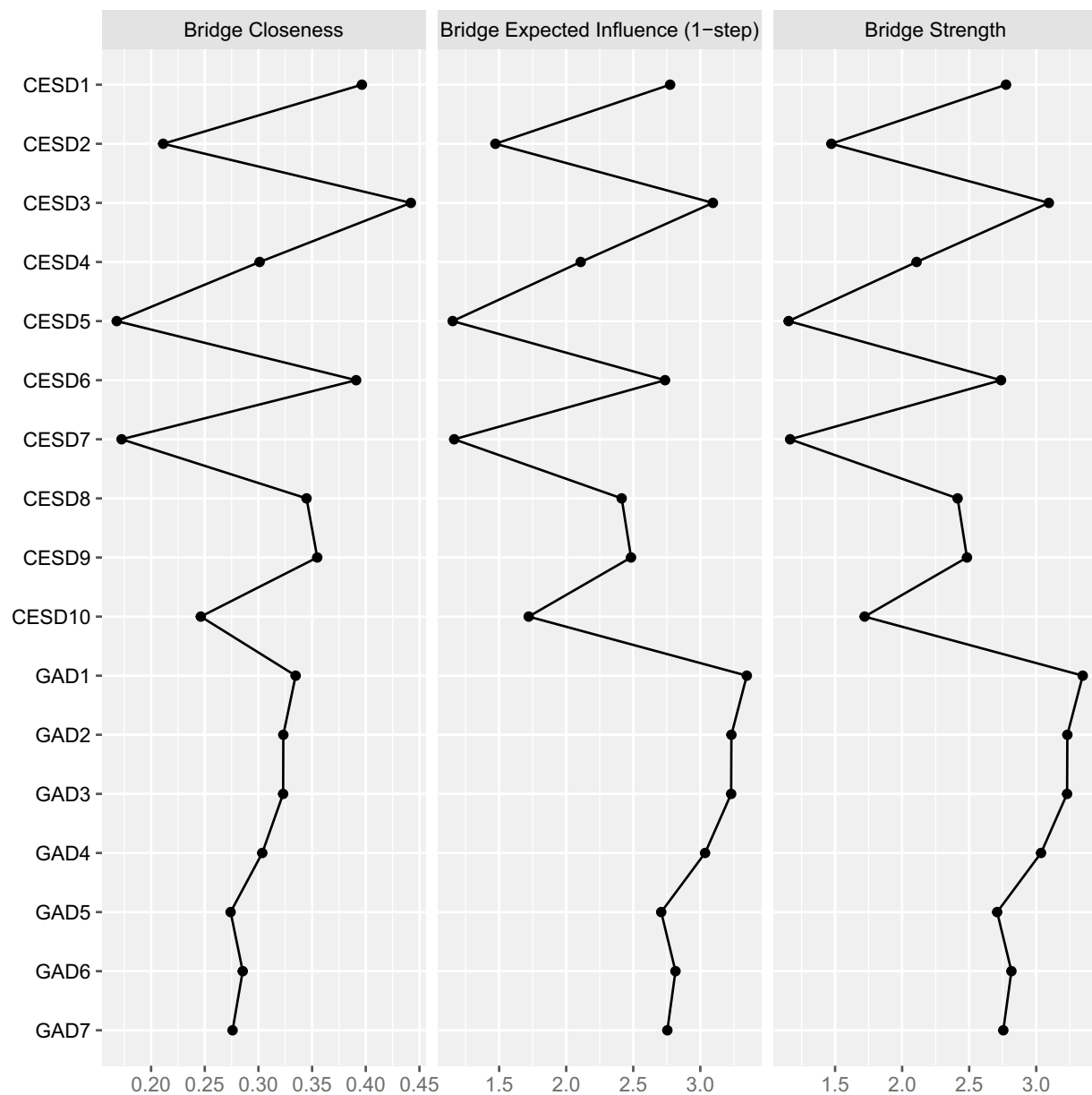


Fig. 3 Bridge symptom analysis in the depression-anxiety network among elderly living alone

(Nervousness or anxiety), GAD2 (Uncontrollable worry), and CESD-3 (Feeling depressed) may represent critical symptoms that contribute to the comorbidity of anxiety and depression among elderly individuals living independently, thereby necessitating further investigation.

From a clinical practice perspective, this study provides significant insights for mental health interventions targeting elderly individuals living alone in China. Intervention strategies should prioritize GAD2 and GAD4, while also considering the bridging role of CESD3 to develop a multidimensional approach to symptom management.

For instance, employing methods such as cognitive behavioral therapy could alleviate anxiety and depression [46–48], thereby enhancing emotional well-being and life satisfaction. Social support is instrumental in alleviating depressive and anxious symptoms among older adults [49, 50]. It is imperative to augment community engagement initiatives for seniors living independently, thereby facilitating their social interactions and diminishing feelings of isolation. Furthermore, it is crucial to consider gender differences, as this awareness can enhance community participation, expand mental health resources,

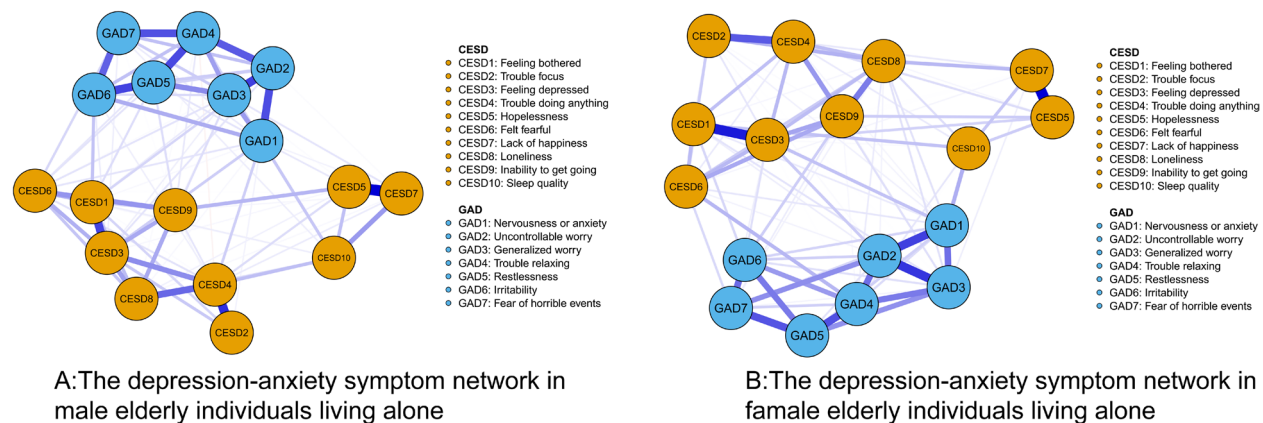


Fig. 4 Depression-anxiety symptom networks in elderly individuals living alone by gender

Table 3 Standardized centrality indices by gender

Label	Items	Male			Female		
		Strength	Closeness	Betweenness	Strength	Closeness	Betweenness
CESD1	Feeling bothered	0.845	0.004	30	0.797	0.004	12
CESD2	Trouble focus	0.466	0.003	0	0.519	0.003	0
CESD3	Feeling depressed	1.062	0.004	38	1.080	0.004	32
CESD4	Trouble doing anything	1.028	0.004	42	0.861	0.003	18
CESD5	Hopelessness	0.673	0.003	14	0.630	0.003	4
CESD6	Felt fearful	0.720	0.003	4	0.713	0.004	14
CESD7	Lack of happiness	0.621	0.003	16	0.616	0.003	4
CESD8	Loneliness	0.795	0.004	4	0.870	0.004	12
CESD9	Inability to get going	0.717	0.004	22	0.805	0.004	30
CESD10	Sleep quality	0.525	0.004	28	0.470	0.004	22
GAD1	Nervousness or anxiety	0.959	0.004	56	1.060	0.004	34
GAD2	Uncontrollable worry	1.108	0.004	30	1.168	0.004	30
GAD3	Generalized worry	1.040	0.004	0	1.057	0.004	6
GAD4	Trouble relaxing	1.122	0.004	24	1.078	0.004	20
GAD5	Restlessness	0.933	0.003	12	0.943	0.004	6
GAD6	Irritability	0.943	0.004	24	0.885	0.004	20
GAD7	Fear of horrible events	0.770	0.003	4	0.872	0.003	6

Note: The numbers in the table represent standardized Z values

and reinforce policy support [51]. Ultimately, these initiatives aspire to improve mental health outcomes and overall quality of life for this vulnerable population. Furthermore, when devising strategies, it is imperative to take into account the cultural disparities between Chinese and Western populations. Considering the heightened dependence of elderly individuals in China on familial support, intervention strategies should prioritize the familial role while alleviating the adverse effects of social pressures on mental health, particularly concerning living arrangements [52]. The government must enhance policy support for mental health by establishing

psychological counseling services and promoting mental health education to combat stigma. Future interventions should embrace a holistic approach that integrates psychotherapy [53], pharmacotherapy [54], and social support [50], while also improving early screening mechanisms to avert the exacerbation of mental health issues among seniors living independently. In light of these considerations, collaboration among government entities, community organizations, and families will be pivotal in enhancing the mental health of elderly individuals living alone, mitigating feelings of loneliness, and improving their overall quality of life.

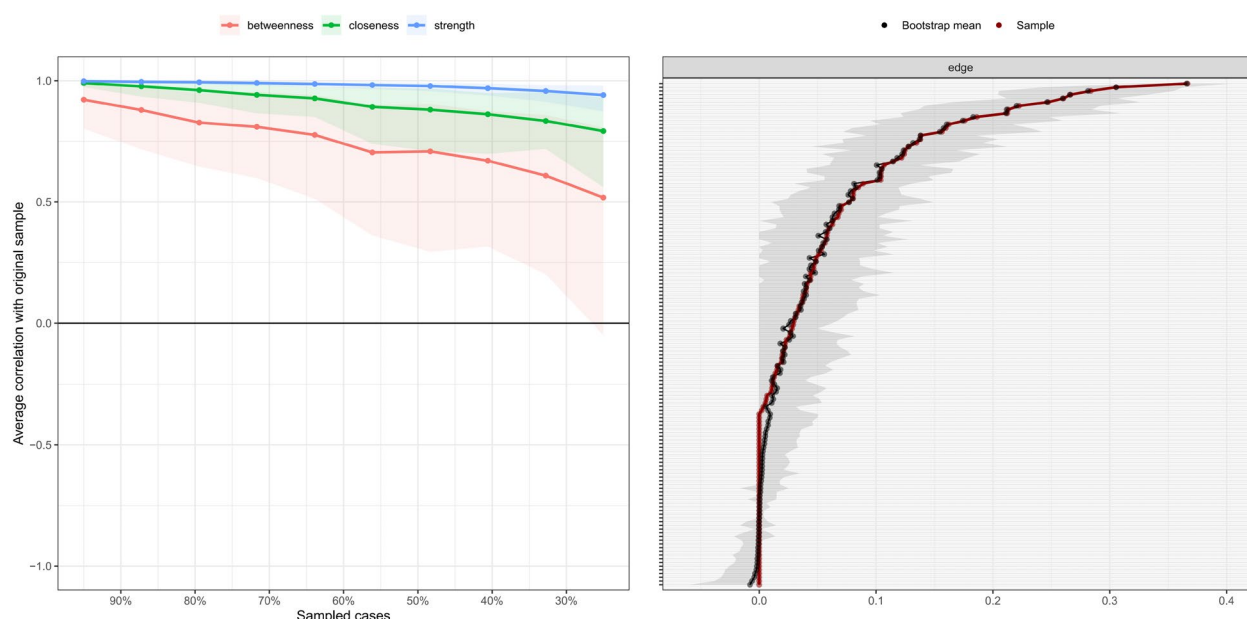


Fig. 5 **A** Stability analysis of the depression-anxiety symptom network in elderly individuals living alone. **B** Accuracy analysis of the edge weights in the depression-anxiety symptom network for elderly individuals living alone (Black lines represent the bootstrapped mean edge weights, while red lines represent the edge weights in the study sample)

Despite the valuable insights provided by this study through network analysis, several limitations should be acknowledged. First, the sample is limited to Chinese elderly individuals living alone, which may limit the generalizability of the findings to other cultural contexts or age groups. Second, the data were collected in 2018, which may not fully reflect the current social environment or the evolving mental health status of elderly individuals living alone. Additionally, the cross-sectional design of this study limits the ability to infer causal relationships. Future research should consider longitudinal designs to explore the dynamic changes in symptom networks over time and the potential causal relationships. Lastly, this study relied on self-reported measures of depression and anxiety symptoms, which may be subject to recall bias and social desirability effects. Future studies could incorporate biomarkers or objective psychological assessments to provide a more comprehensive understanding.

Conclusions

In summary, This study highlights the critical features of anxiety and depression symptom networks in older adults living alone, identifying GAD2 (Uncontrollable worry) and GAD4 (Trouble relaxing) as the core symptoms. Targeted interventions focusing on these central nodes may effectively disrupt the symptom network's vicious cycles and alleviate the overall symptom burden in this population. Future research should validate

these findings in clinical settings to support precision treatment for anxiety and depression in older adults living alone.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12888-024-06443-2>.

Supplementary Material 1

Acknowledgements

Not applicable.

Authors' contributions

Yunling Zhang and Lina Miao conceived and designed the research. Ze Chang and Yunfan Zhang performed the research and revised the manuscript. Xiao Liang, Yunmeng Chen, Chunyan Guo, Xiansu Chi, Liuding Wang, Xie Wang, Hong Chen, Longtao Liu, and Zixuan Zhang analyzed the data and draw figures.

Funding

This research was supported and funded by the Scientific and Technological Innovation Project of CACMS (No. CI2021B006), CACMS Innovation Fund (No. CI2021A01301, CI2021A01311), Hospital capability enhancement project of Xiyuan Hospital, CACMS (No. XYZX0204-05, XYZX0101-21), the Innovation Team and Talents Cultivation Program of National Administration of Traditional Chinese Medicine (No. ZYYCXTD-C-202007), the China Postdoctoral Researchers National Funding Program Project (GZC20233131), the National TCM Leading Personnel Support Program (NATCM Personnel and Education Department (2018) (No. 12).

Data availability

The data for this study were sourced from the Chinese Longitudinal Healthy Longevity Survey and Family Happiness (CLHLS-HF), conducted by the Peking University Research Center for Healthy Aging and Development. We have also obtained authorization to use this data for our research.

Declarations

Ethics approval and consent to participate

The CLHLS study received ethical approval from the Peking University Institutional Review Board (IRB00001052-13074), and written informed consent was obtained from each participant or their authorized representative. Participation in the CLHLS is voluntary, and respondents are free to withdraw from the study at any time without any consequences.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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Received: 1 September 2024 Accepted: 23 December 2024

Published online: 08 January 2025

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